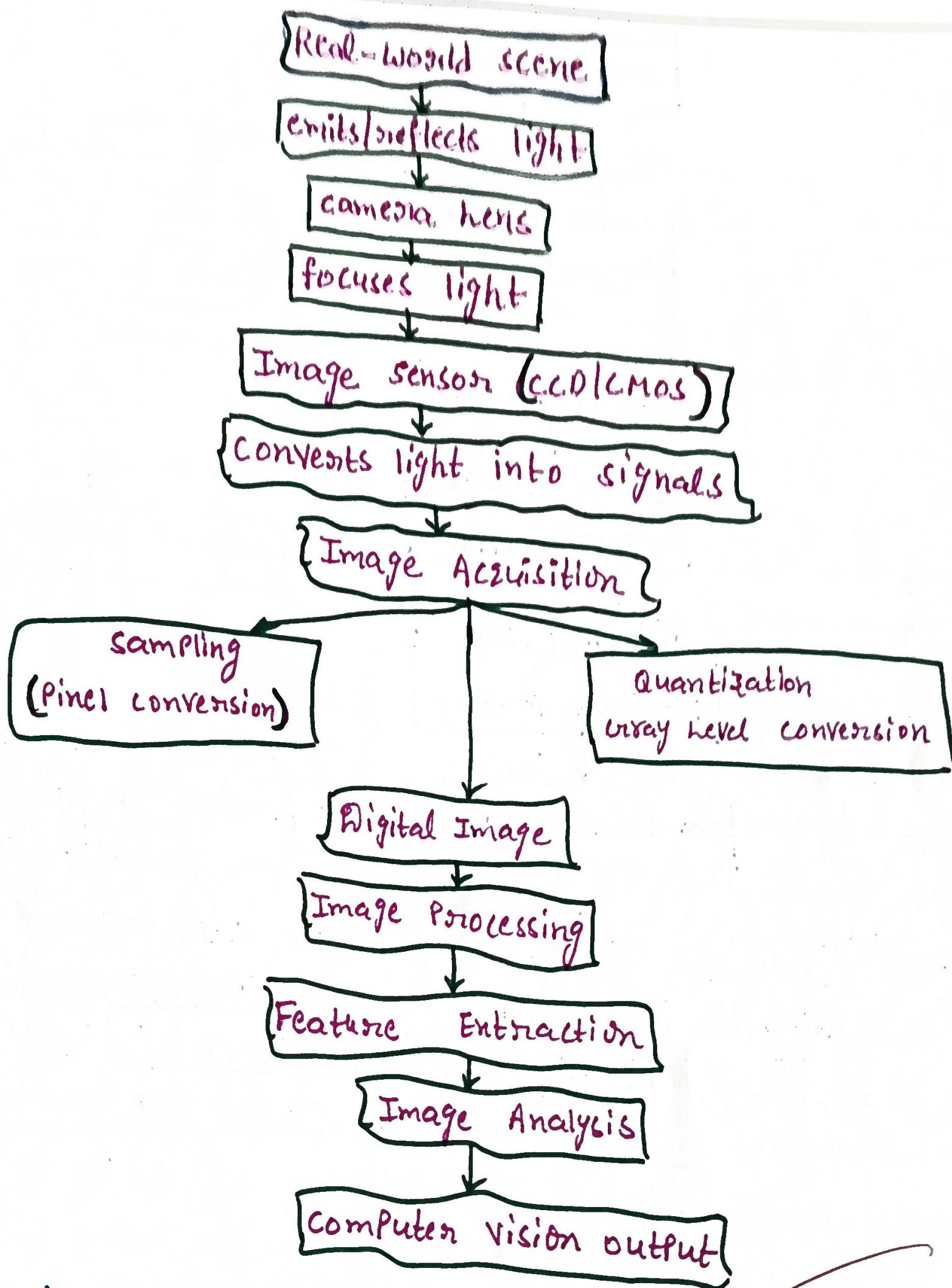
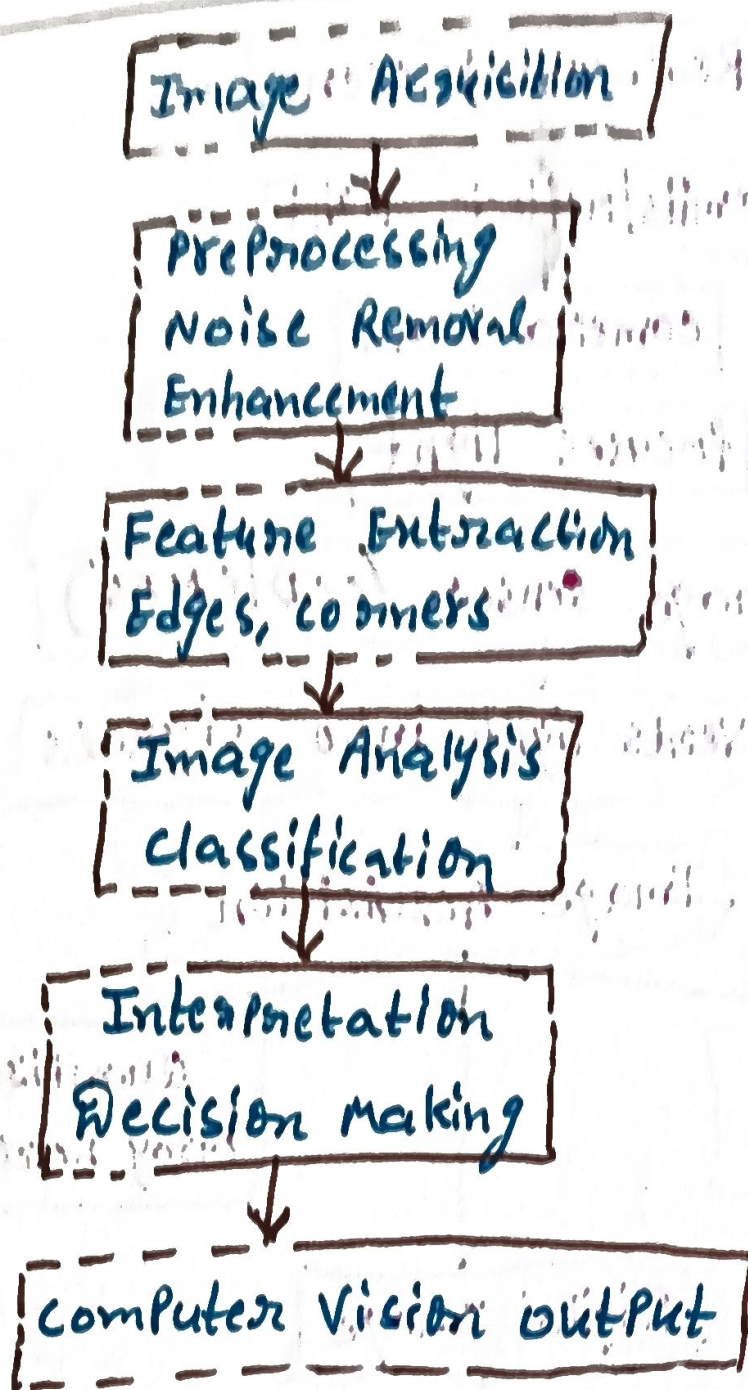




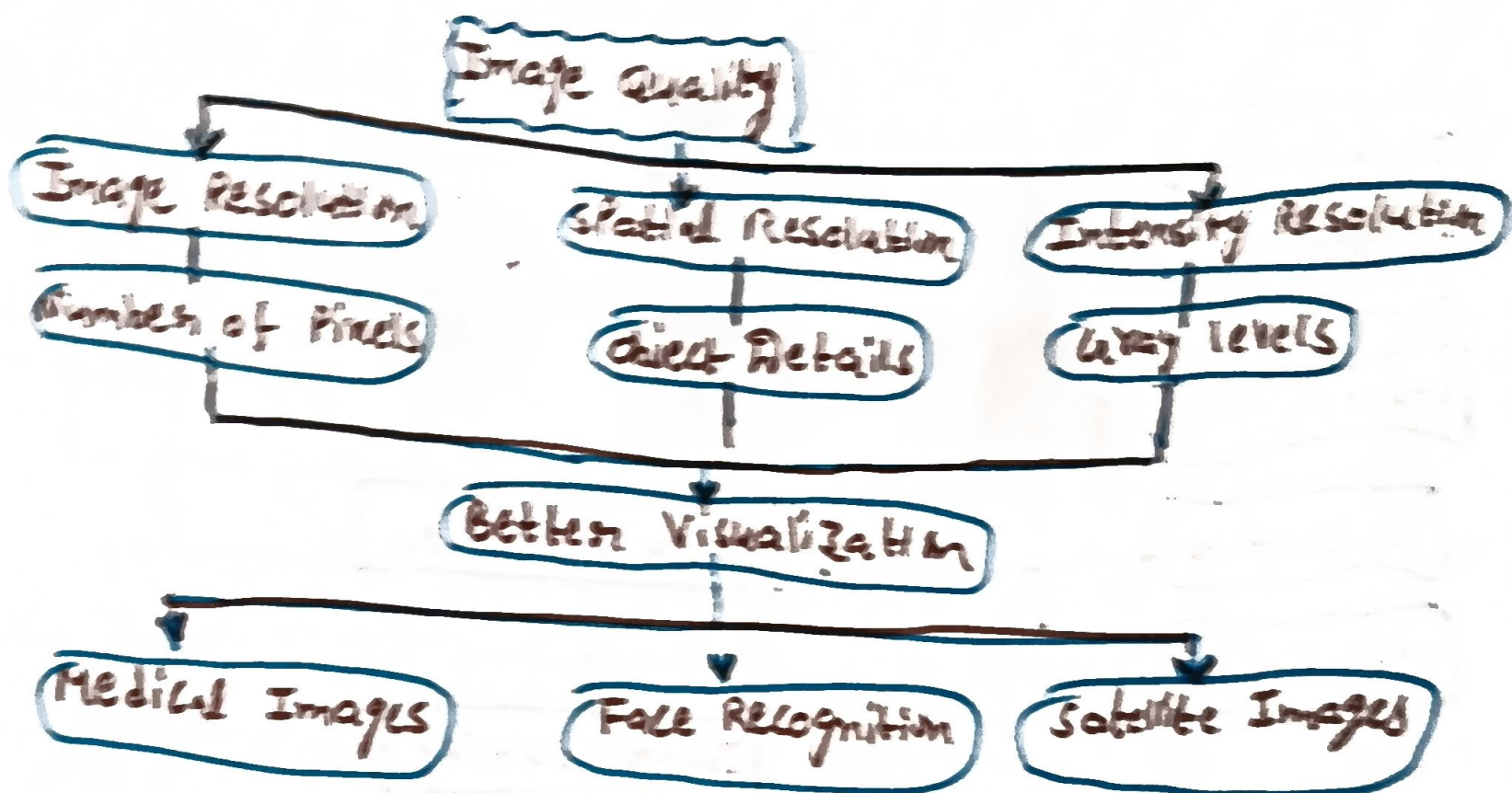
- A real world scene is captured using a camera. The image sensor converts light into electrical signals, which are transformed into a digital image through sampling and Quantization. Image Processing and feature extraction enable the computer vision system to analyze the image and produce useful results.



- The computer vision system first captures an image and improves its quality using preprocessing. Important features are extracted, analyzed, and interpreted to make meaningful decisions such as object recognition and classification.

References

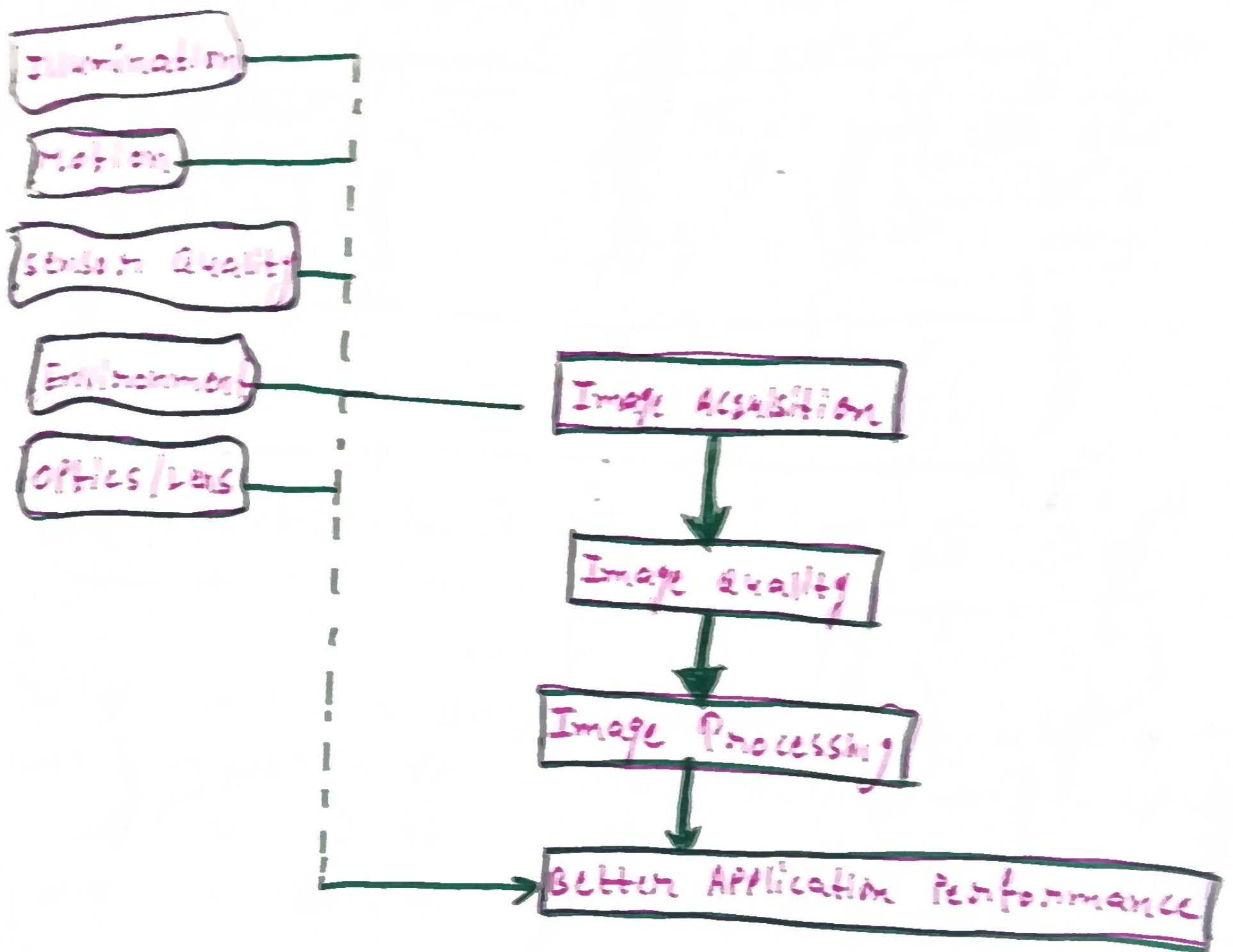
- Szeliski (2022)
- Gonzalez & Woods (2023)
- OpenCV Documentation (2024)
- IEEE Transactions on PAMI (2024)
- Elsevier Computer Vision Review (2025)



• Image quality depends on image resolution, spatial resolution, and intensity resolution. Higher resolution provides more image details and better contrast, making the image suitable for medical, satellite and surveillance applications.

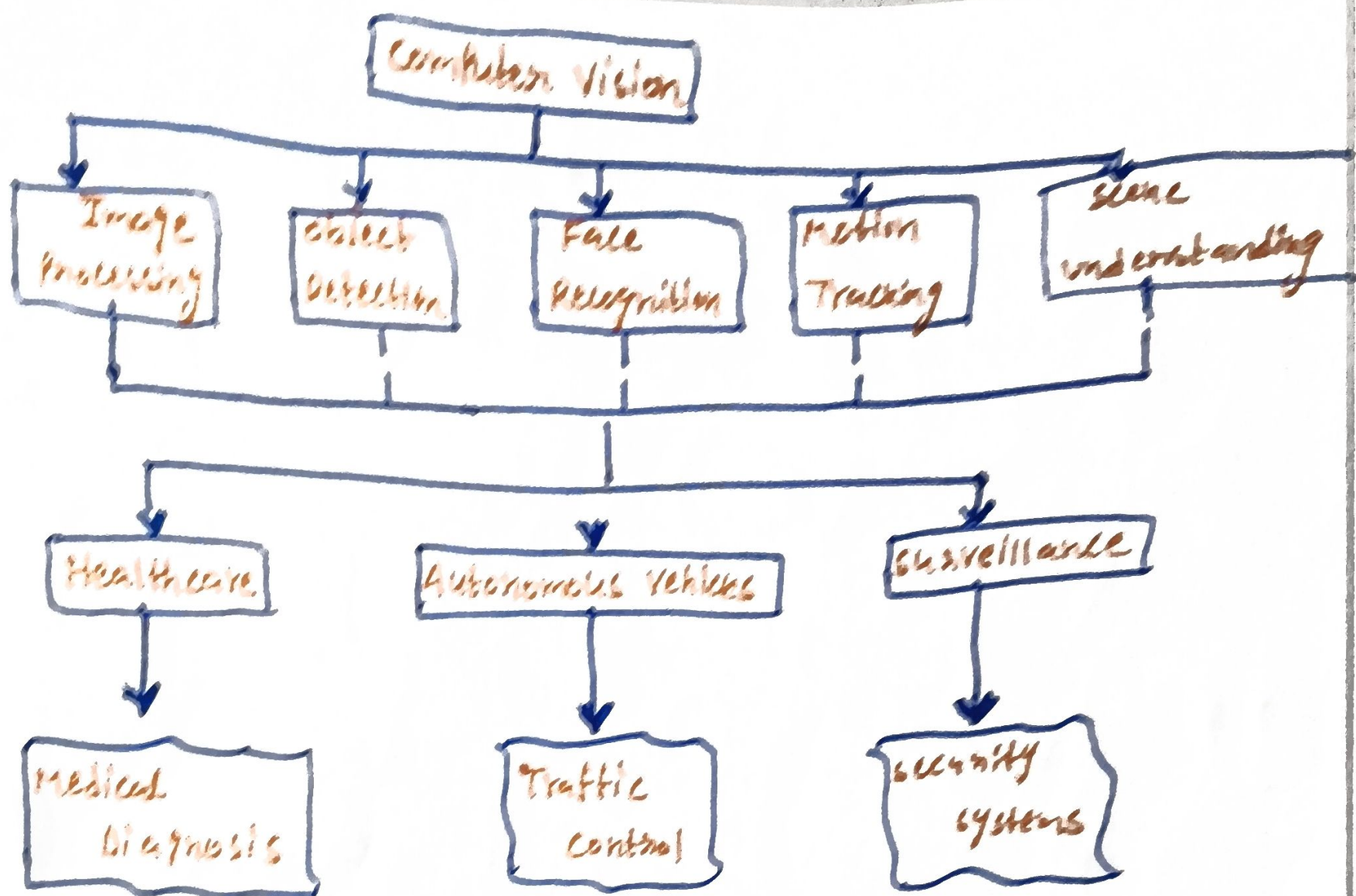
References:

- 1) Gonzalez & Woods (2023)
- 2) Spring Image Processing (2023)
- 3) IEEE Image Processing Journal (2024)
- 4) Open CV Documentation (2024)
- 5) Elsevier Digital Imaging Reviews (2025)



Explanation :

Image acquisition is affected by lighting, camera sensors, object motion, environmental conditions, and lens quality. These factors influence image quality and the Performance of computer vision applications.



Explanation:

Computer Vision provides capabilities such as image processing, object detection, face recognition, motion tracking, and scene understanding. These capabilities are widely used in healthcare, surveillance, robotics, agriculture, and autonomous vehicles.